VP Vidyalankar Polytechnic	Electronics and Computer Engineering

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Batch	**3**
Subject name	**Python Programming & Data Structure**
Teacher Name	**Ms. Rasika Patil**

Experiments	**2**
Title	**a) Develop python program to check whether the entered frequency is radio frequency or audio frequency.** **b) Develop python program to display various radio frequency bands using if..elseif ladder.** **c) Develop python program to display resistor color code using switch statement.**

Theory

Decision-making in Python uses conditional statements such as if, elif, and else to control program flow.

In this practical, students classify frequencies as **Audio Frequency (AF)** or **Radio Frequency (RF)**, identify RF bands, and implement resistor color code logic.

Syntax

• if condition:
• elif condition:
• else:
• Comparison operators: < > == != >= <=

Simple If Statement: A simple `if` statement is used to execute a block of code only if a specified condition is true. If the condition is false, the block of code inside the `if` statement is skipped.

Syntax:

if condition:

code to execute if condition is true

Example:

```
age = 18

if age >= 18:

    print("You are eligible to vote.")
```

In this example, the message "You are eligible to vote." is printed only if the `age` is 18 or older. **If-Else Statement**: An `if-else` statement is used to execute one block of code if the condition is true, and another block of code if the condition is false.

Syntax:	Example:

if condition:
 	
# code to execute if condition is true 	
else:
 	
# code to execute if condition is false 	
	age = 16
	
	if age >= 18:
	
	print("You are eligible to vote.")
	else:
	
	print("You are not eligible to vote.")

In this example, the message "You are not eligible to vote." is printed because the `age` is less than 18.	

Nested If Statement

A nested `if` statement is an `if` statement inside another `if` statement. This allows for more complex decision-making and multi-level condition checking.

Syntax:	Example:

if condition1:
 	
# code to execute if condition1 is true if condition2: 	
 	
# code to execute if both condition1 and condition2 are true 	
	age = 20
	has_id =
	True if age
	>= 18:
	
	if has_id:
	
	print("You are eligible to enter the club.")
	else:
	
	print("You need an ID to enter the club.") else:
	
	print("You are not old enough to enter the club.")

In this example, the message "You are eligible to enter the club." is printed because both conditions (`age >= 18` and `has_id`) are true. If `has_id` were `False`, the message "You need an ID to enter the club." would be printed instead.	

Programming Questions with Answer	
1	**Write a program to check whether entered frequency isradio frequency or audio frequency.**
SCRIPT	f=float(input ("Enter frequency in Hz:")) if f <=2000: print("Audio frequency(AF)")

	else: print("Radio frequency(RF)")
OP	Enter frequency in Hz: 200 Audio frequency(AF)
2	**Program to display RF bands using if–elif**

SCRIPT	f=float(input("Enter frequency in Hz:"))
	if 30000 <=f<=300000:
	print("VLF Band")
	elif 300000<=f<=3000000:
	print("LF Band")
	elif 3000000 <=f<=30000000:
	print("HF Band")
	elif 30000000 <=f<=300000000:
	print("VHF Band")
	else:
	print("Frequency not in standard RF range")

OP	Enter frequency in Hz: 330000 LF Band
3	**Program for resistor color code using selection logic**

SCRIPT	color=input("Enter color:").lower()
	if color=="black":
	print("Value=0")
	elif color=="brown":
	print("Value=1")
	elif color=="red":
	print("Value=2")
	elif color=="orange":
	print("Value=3")
	elif color=="yellow":
	print("Value=4")
	elif color=="green":
	print("Value=5")
	elif color=="blue":
	print("Value=6")
	elif color=="violet":
	print("Value=7")
	elif color=="gray":
	print("Value=8")
	elif color=="white":
	print("Value=9")
	else:
	print("Invalid color")

OP	`Enter color: white` `Value=9`
4	**Write a Python program that checks if the voltage of a battery is greater than 3.7V and prints "Battery is charged" if true.**

SCRIPT	voltage=3.7
	if voltage==3.7:
	  print("Battery is charged")
	else:
	  print("Battery is not charged")

OP	`Battery is charged`
5	**Write a Python program that checks if a resistor value is within a tolerance range (e.g., 5% tolerance for a 100Ω resistor) and prints whether it is within tolerance or not.**

SCRIPT	nominal_resistance = 100
	tolerance_percent = 5
	measured_resistance = 104
	
	tolerance = nominal_resistance * (tolerance_percent / 100)
	lower_limit = nominal_resistance - tolerance
	upper_limit = nominal_resistance + tolerance
	
	if lower_limit <= measured_resistance <= upper_limit:
	  print("The resistor is within tolerance.")
	else:
	  print("The resistor is out of tolerance.")

OP	`The resistor is within tolerance.`

1	**Debugging Questions with Correct:**
	f = float(input("Frequency: "))
	if f < 20000
	  print("AF")
	else
	  print("RF")

OP	**Error:** : is required after if and else. **Correct code :f = float(input("Frequency: "))** **if f < 20000:** **print("AF")** **else:** **print("RF")** Frequency: 500 AF
2	**Debug the following code** **Wrong code:** clr = input("Color: ") if clr = "red": print(2)
OP	**Error:** == checks whether two values are equal **Correct code:** **clr = input("Color: ")** **if clr == "red":** **print(2)** Color: red 2
Conclusion	These programs help identify different frequency types and radio bands using conditional logic in Python. The switch (match-case) structure makes color code selection simple and clear. Overall, the programs show how Python can be used to classify and display technical data effectively.

Performance Indicators		**Teachers Dated and Signature**
Process Related (15 M)	**Product Related (10 M)**	**Total (25 M)**